Homework 3

Submission 2, Spring 2026

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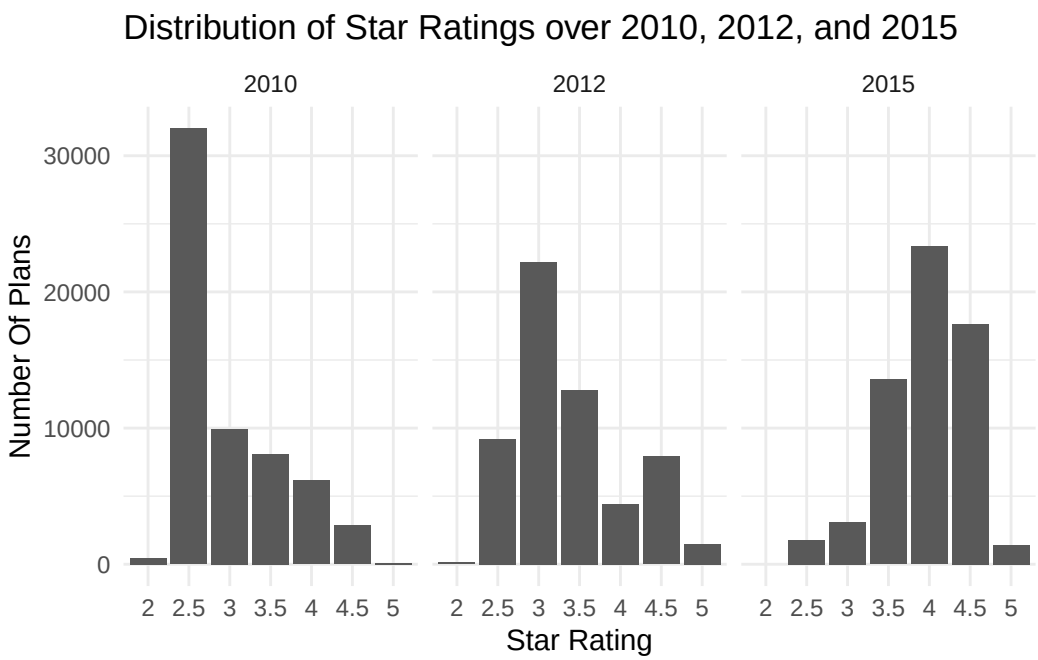
Problem 1

```
# A tibble: 6 x 5
   year `Average Star Rating` `Average Enrollments` `Average Market Share`
  <int>                 <dbl>                 <dbl>                  <dbl>
1  2010                  2.97                  256.                 0.0658
2  2011                  3.26                  343.                 0.0855
3  2012                  3.36                  377.                 0.0856
4  2013                  3.55                  396.                 0.0815
5  2014                  3.81                  437.                 0.08
6  2015                  3.96                  470.                 0.0787
# i 1 more variable: `Total Number Of Plans` <int>
```

Problem 2

```
# A tibble: 6 x 4
   year `Average Enrollments` `Average Market Share` `Total Number Of Plans`
  <int>                 <dbl>                  <dbl>                   <int>
1  2010                  107.                 0.0673                   48643
2  2011                  168.                 0.09                     12754
3  2012                  173.                 0.0567                    9077
4  2013                  214.                 0.06                      3661
5  2014                  258.                 0.0557                    3708
6  2015                  314.                 0.0405                    4680
```

Problem 3



We note that the there is steep decrease in the number of high-rated plans in 2010, whereas in 2012 the difference is less stark, and in 2015, we notice an improvement, with existence of a larger number of high-rated plans.

Problem 4

```
                          2010    2011    2012   2013   2014   2015
Excluded (<= 2.5 stars)  0.0634  0.0702  0.0789 0.0491 0.0640 0.0606
3 Stars                  0.0126  0.0350  0.0307 0.0621 0.0267 0.0271
3.5 Stars                0.0005  0.0191  0.0129 0.0385 0.0273 0.0388
4 Stars                 -0.0036  0.0028 -0.0246 0.0172 0.0074 0.0115
>= 4.5 Stars            -0.0076 -0.0095 -0.0134 0.0222 0.0071 0.0233
```

Problem 5

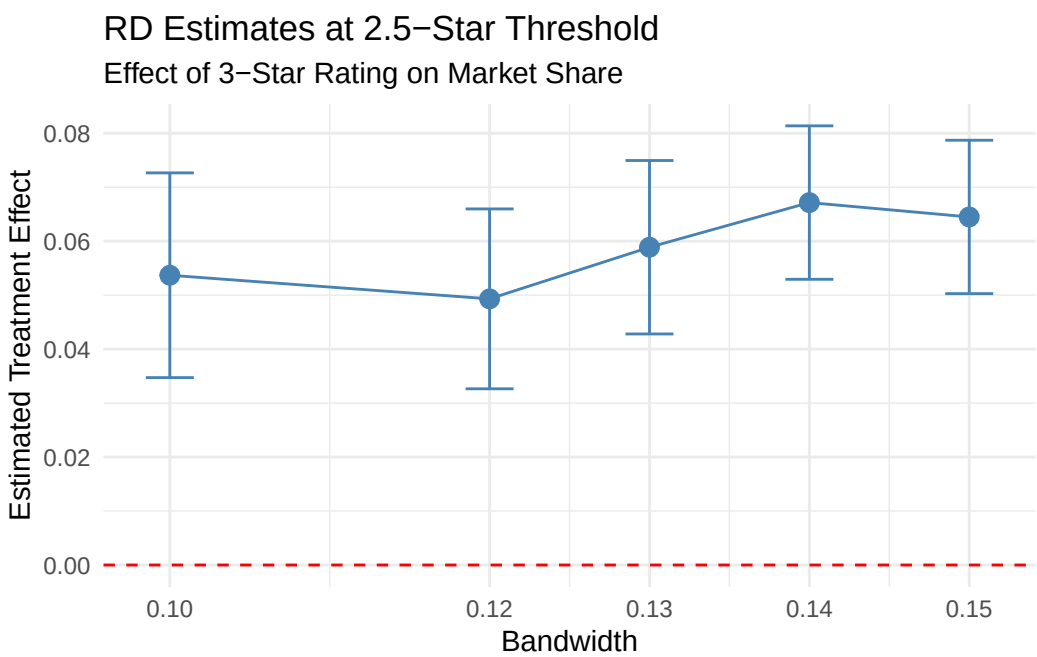
```
# A tibble: 5 x 2
  `Star Rating`     `Number Of Plans`
  <chr>                         <int>
1 Rounded up to 3                1978
2 Rounded up to 3.5              2073
3 Rounded up to 4                1056
```

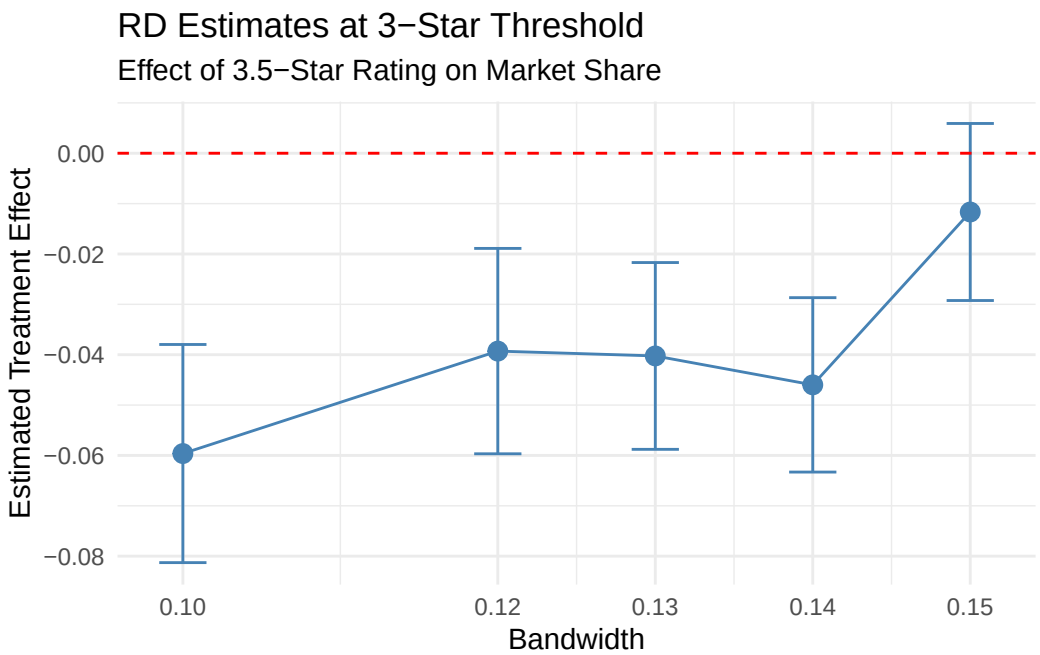
```
4 Rounded up to 4.5                150
5 Rounded up to 5                   15
```

Problem 6

```
            X3.vs.3.5   X2.5.vs.3
RD Estimate 0.0546005 -0.03953024
```

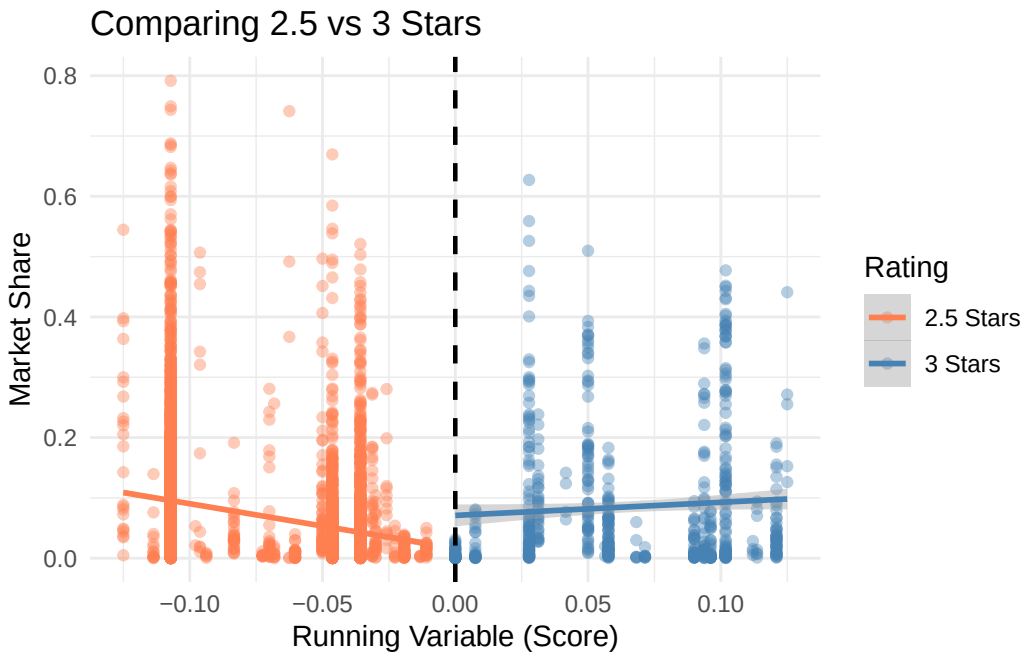
Problem 7

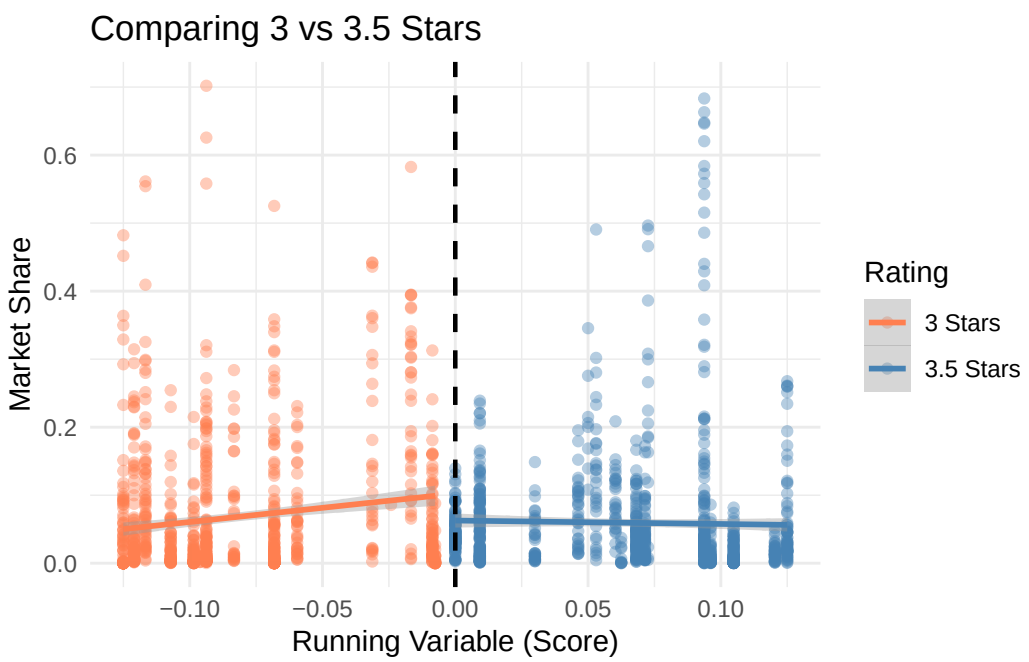




We note that findings are indeed sensitive to bandwidth choice. For instance, the treatment effect in comparing in 3 versus 3.5 star plans appear to be more vivid when compared at the threshold of 0.1 as opposed to 0.15. Similarly, we see a difference in calculated effect when a bandwidth of 0.12 is chosen as opposed to 0.14 while comparsing 2.5 star plans versus 3 star plans.

Problem 8





We do see an abrupt increment in market share concording to values below and above the threshold, particularly prominent near the threshold of comparison for 2.5 stars against 3 stars (i.e., the first plot). Particularly, the drop from -0.10 to -0.05 in the upper plot and a slight increase from 0.05 to 0.10 in the lower plot.

Problem 9

```
       Threshold PartD_Yes  PartD_No Below_Threshold Above_Threshold
1 2.5 vs 3 Stars 0.5581264 0.4418736       0.9082487      0.09175132
2 3 vs 3.5 Stars 0.8783270 0.1216730       0.5661778      0.43382220
```

With a bandwidth of 0.125, we note that there is a substantially higher proportion of below-threshold contracts than above, for both categories. The threshold was calculated via score, i.e, adjusted raw rating, as HMO is constituted through Part C and defined here as score. Furthermore, we see the proportion of Part D plans significantly increasing from 55% to 87%, going from 3-star to 3.5-star comparison. Thus, there are indeed varying characteristics between plans.

Problem 10

The effect of increased star rating is directly proportional to a larger number of enrollments, particularly prominent from the upward and downward slopes, above and below the threshold, respectively, in the Problem 8 plot comparing 3-star plans against 3.5-star ones. However, these results are highly sensitive to the chosen bandwidth (Problems 6 & 7); regardless, we see that it brings an incremental distribution of the number of plan D contracts going from 3-star threshold to 3.5-star threshold (Problem 9). Further, an effect is added from the fact that the 3-star ratings experience the largest number of round-ups as compared to plans above it, thereby making that region particularly volatile to analysis.